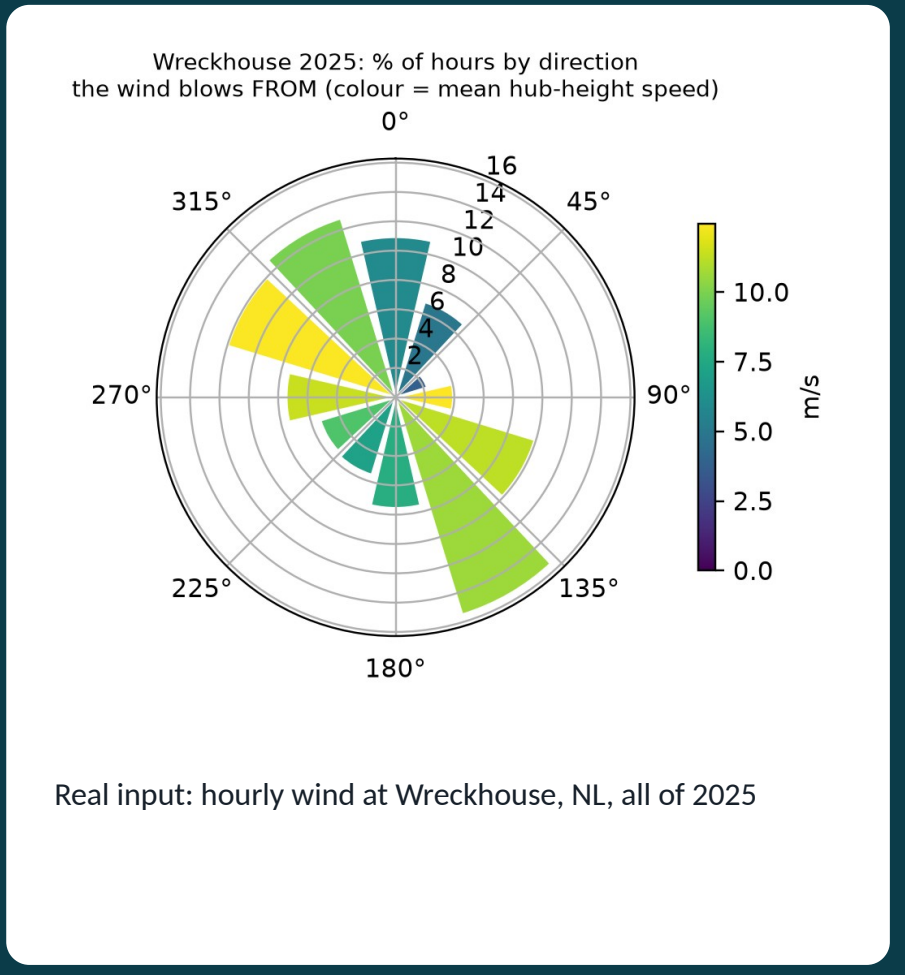
WakeWise NL

Wake-aware wind-turbine siting for Newfoundland's west coast, solved with QAOA in Qiskit and run on IBM Quantum System One in Quebec

Team AI Entanglers

Qiskit Fall Fest 2026 · Memorial University



Real input: hourly wind at Wreckhouse, NL, all of 2025

WakeWise NL in one minute

180 GWh/yr

from the best 10-turbine layout on our 12-site Wreckhouse parcel: the electricity of ~10,600 NL homes

~380x

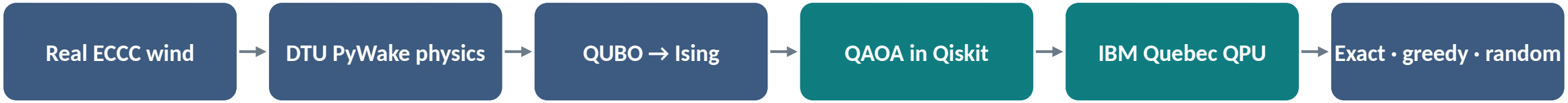
more often than random guessing, QAOA returns the optimal layout (12 qubits, noiseless simulation)

68x

random on real hardware: ibm_quebec finds the 12-site optimum in 1.7% of shots

0.99

rank correlation between our QUBO and full DTU PyWake wind-farm simulations: the model we optimize is the right one



Teal = quantum steps. No quantum advantage is claimed: at this size we can check every answer exactly.

Strong NL wind meets a proposed hydrogen farm

Wreckhouse, southwest Newfoundland

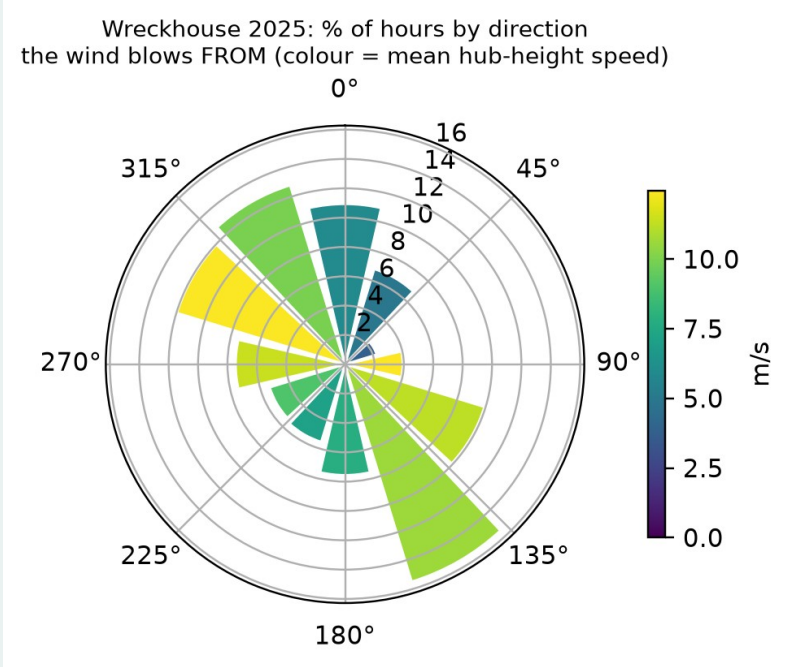
Gales here once blew trains off the tracks. In 2025 the wind at a 110 m hub averaged 9.4 m/s: a 4 MW turbine would run at a 53% capacity factor.

Project Nujio'qonik, next door

150+ turbines proposed on the Port au Port Peninsula and in the Codroy Valley to make hydrogen in Stephenville. Released from provincial environmental assessment, with conditions, in April 2024.

MUN's CREATE-SWIFT program asks for this

Benchmark wind-energy optimization: classical versus quantum or hybrid methods, on accuracy, cost, scalability and real-world relevance.



Bimodal wind: south-easterly gales and north-westerlies. Turbines lined up on that axis shade each other most.

Sources: ECCC hourly climate data (Wreckhouse, ID 8404343); NL Government EA bulletin, 9 Apr 2024; CBC News; CREATE-SWIFT talk, Qiskit Fall Fest 2026.

Wakes waste wind, so siting is a real decision

18,500 MWh/yr

from one 4 MW turbine standing alone at Wreckhouse (real 2025 wind, DTU PyWake)

up to 1.9%

of that lost to one badly placed neighbour's wake; a dense 12-turbine block loses 3.6%

35% of hours

at full power, when a wake costs almost nothing: why Wreckhouse wakes are milder than at an average site

Decision and intended use

Which candidate sites get a turbine (bit = 1) or not (bit = 0). Output: the best layouts and the energy at stake, for early-stage screening, not construction advice.

Sustainability benefit

More clean electricity, or green hydrogen, from the same land, steel and concrete; or the same energy from fewer turbines and less disturbed ground. The prototype shows the method; real benefit needs site-specific terrain, turbines and costs.

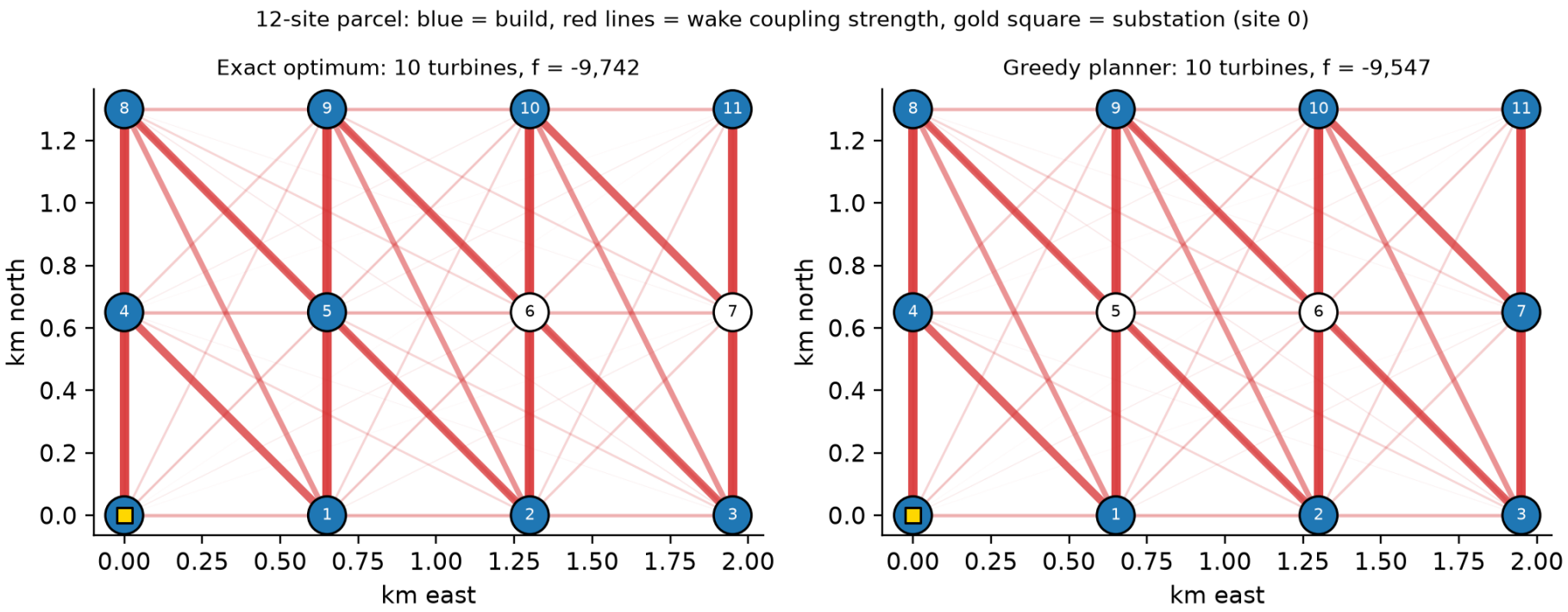
Inputs, output and scope

Input	Value used	Status
Hourly wind direction and speed	Wreckhouse, ECCC, 2025 (8,740 hours)	**Observed**
Hub-height wind	10 m to 110 m with power-law shear, alpha = 0.14	Assumption
Wind-farm physics	DTU PyWake: Jensen wake model; checked with full-farm and Gaussian models	**Industry-standard model**
Turbine	4 MW, 130 m rotor, power and thrust curves from PyWake's generic model	Illustrative class
Parcel	Flat grid, 650 m (5 rotor diameters) apart	Illustrative
Costs, as energy	88% of a turbine's output + 2% per 650 m of cable	Illustrative, sensitivity tested
Output	Build / no-build bitstring = one layout	
Scope	9 and 12 sites (9 and 12 qubits): simulation, IBM noise model and real hardware	

Wake interactions make the choice combinatorial

$$f(x) = -\Sigma\,(E - c_i)\,x_i \;+\; \Sigma\,W_{ij}\,x_i\,x_j$$

E: one turbine's energy · c_i: annualized cost as energy · W_{ij}: energy the pair loses to each other's wakes (PyWake) · MWh/yr, lower is better



Baselines, same instance

- Exact search of every layout
- Greedy: add the best site until none helps
- Uniform random guessing (control)

Greedy is optimal on 9 sites and 195 MWh/yr short on 12

The wake physics is already an Ising model

$$x_i = (1 - z_i) / 2$$

$$H = \Sigma\, h_i Z_i + \Sigma\, J_{ij} Z_i Z_j$$

Turbine value becomes a Z field; the wake loss between two turbines becomes a ZZ coupling.

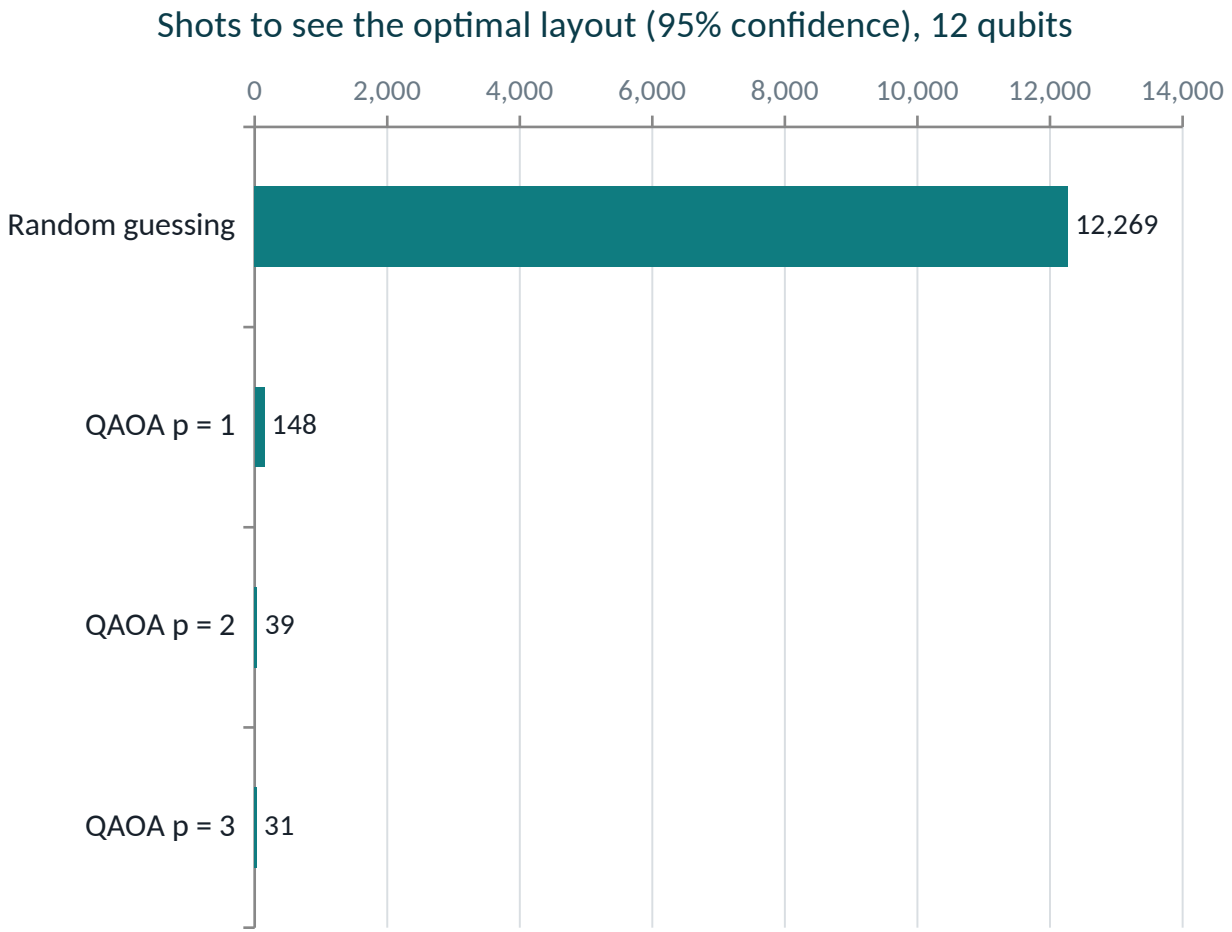
Why QAOA fits

- Natively quadratic and binary: no encoding overhead, no penalty terms
- NP-hard in general (contains maximum-weight independent set)
- CVaR training: optimize the best 25% of outcomes, because a planner needs the best layout
- Small enough to check exactly, so quality is measured, not assumed

1. **Map** *classical*
   ECCC wind → PyWake → QUBO → SparsePauliOp → qaoa_ansatz, depth p = 1 to 3
2. **Optimize** *classical*
   Transpile for IBM Heron (level 3); drop weak couplings to save two-qubit gates
3. **Execute** *quantum*
   Train angles on exact simulation (CVaR + COBYLA); sample layouts: ideal, noise model, ibm_quebec
4. **Post-process** *classical*
   Score every sampled layout with the full QUBO; compare with exact, greedy and random

QAOA finds the best layout ~380x more than chance

Noiseless simulation



9.2% vs 0.02%

of shots return the optimal layout (QAOA p = 3 vs random guessing)

31 vs 12,269

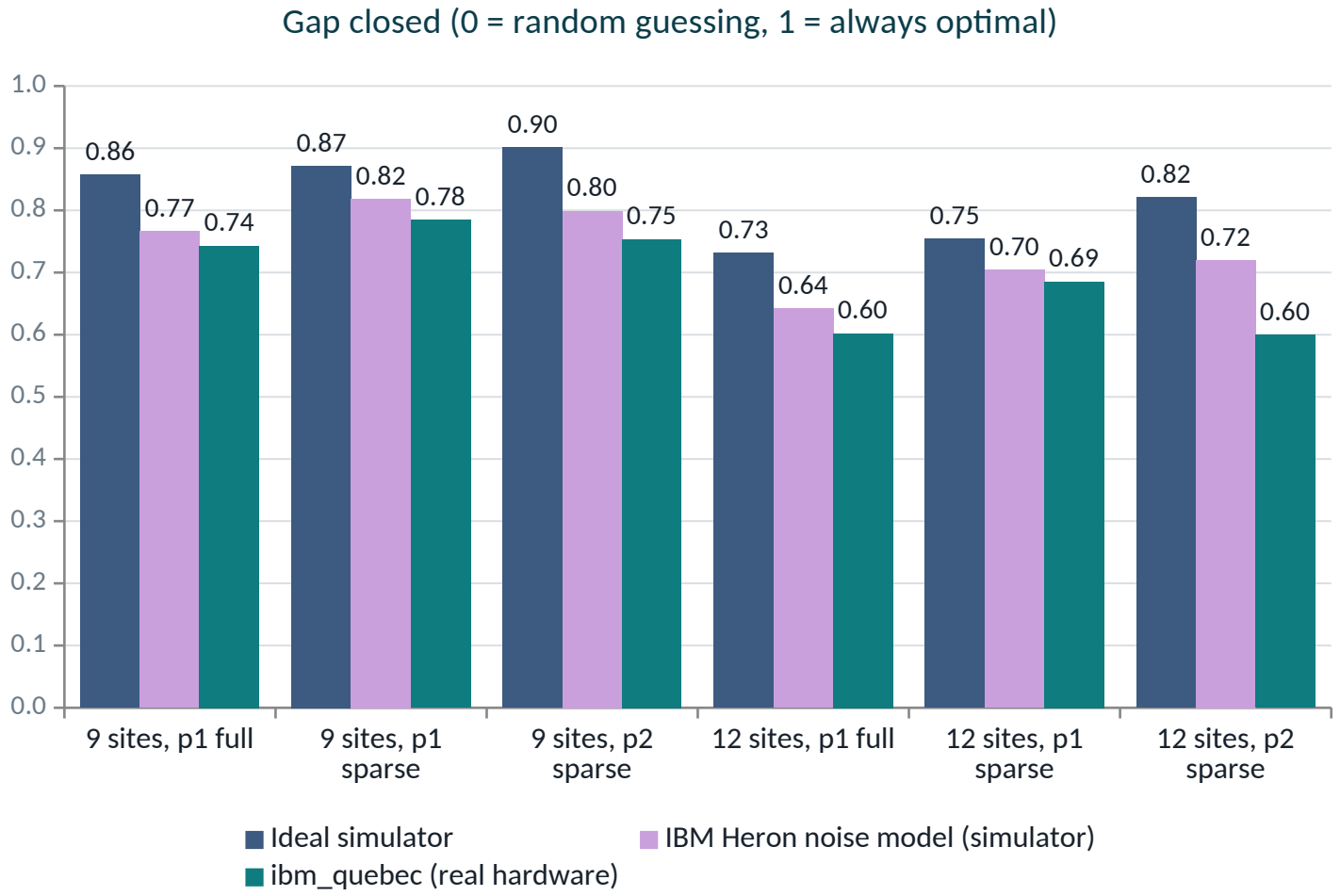
shots needed to see the optimum with 95% confidence

9.2% vs 8.3%

CVaR training vs standard energy training at p = 3

Honest caveat: a typical QAOA sample (gap closed 0.84) is still below greedy's single answer (0.94). QAOA's value is in its best few dozen shots.

On IBM's Quebec machine, the signal survives



Simulator + real hardware

ibm_quebec (IBM Quantum System One at PINQ², Bromont, Quebec), 7 Nov 2026; median CZ error 0.15%.

12 qubits: best circuit (depth 2, sparsified) finds the optimum in 1.7% of shots, 68x random.

9 qubits: 10.3% of shots, 53x random.

Control circuit (should be random) scores -0.02: the gains come from QAOA, not from noise.

4,000 shots per circuit; dynamical decoupling and gate twirling; angles trained in simulation, so no QPU time is spent on training.

Our QUBO matches full wind-farm simulations



0.994

rank correlation with full-farm PyWake (all wakes at once)

0.997

rank correlation with a different, Gaussian wake model

#1 and #1

rank of the QUBO's optimal layout under those two simulations

The QUBO adds wake losses pair by pair; a full simulation lets every wake act at once. They agree, so optimizing the QUBO optimizes the farm.

What the evidence supports, and what it does not

Supports

- Siting is combinatorial on real NL wind; the pairwise QUBO matches full wind-farm simulations
- QAOA biases sampling toward the best layout (~380x random at p = 3) and improves with depth; CVaR training helps
- The signal survives real hardware: ibm_quebec finds the optimum 68x more often than chance on 12 qubits
- Sparsifying weak couplings cuts two-qubit gates with little loss in quality

Does not support

- Quantum advantage: a laptop checks all 4,096 layouts in milliseconds
- Large energy gains: with realistic Wreckhouse wakes greedy is within 2% of optimal
- Real-world energy figures: flat idealized parcel, one wind year, illustrative costs
- QAOA beating greedy on a typical sample: only its best samples do

Limitations and next steps

Limitations and resources

- Model: flat terrain, one wind year, 10 m data extrapolated to hub height, a generic turbine class
- Quantum: depth $p \leq 3$ trained in simulation; dense couplings need SWAPs on heavy-hex, so hardware circuits stay shallow
- Resources: the full notebook runs in a few minutes on a laptop; hardware used two jobs of 4 circuits × 4,000 shots

Next steps

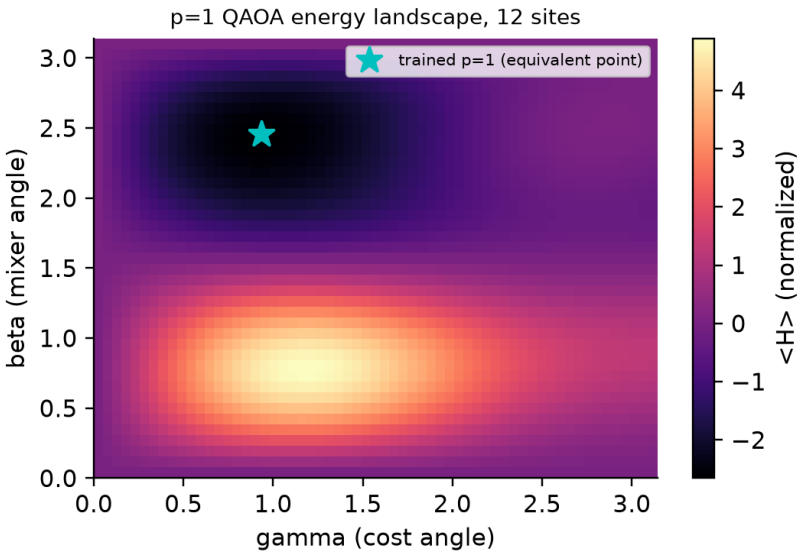
1. Real Codroy Valley parcel with terrain-aware wind
2. Spacing and exclusion zones (wetlands, caribou habitat) as extra QUBO terms
3. Constraint-preserving mixers and warm starts
4. Scale past exact search; compare with simulated annealing and MILP
5. Partner route: MUN CREATE-SWIFT, with Mitacs Accelerate funding

More clean energy from the same wind

- Real Wreckhouse wind + DTU PyWake physics give a QUBO that matches full wind-farm simulations
- QAOA in Qiskit finds the best layout ~380x more often than chance, and 68x on IBM's Quebec machine
- Honest scope: a verified proof of concept, no advantage claimed

github.com/mrw-soumik/Qiskit-AI-Entanglers-

Thank you. Questions?



p = 1 QAOA energy landscape for the 12-site parcel; the star marks the trained angles.

Credits: QAOA workflow adapted from Tinh Bui's workshop notebook; IBM Qiskit 101 notebook and Fall Fest hardware example; DTU PyWake; ECCC data (Open Government Licence). Full credits in the notebook.